AI TRAVEL WATCHDOG & GBV SAFETY COMPANION

Protecting Every Commuter. Zero-Knowledge Safety in 11 Languages.

Phola combines real-time transit anomaly detection, native in-app turn-by-turn navigation to SAPS police stations, 1-tap SOS panic, and encrypted digital evidence preservation.

JUDGE / ASSESSOR TESTING ACCOUNT (LIVE & ACTIVE)

`Email: jackympoka@gmail.com | Password: Mpokabetty221@`

Live Progressive Web App

`https://phola-beta.vercel.app`

PROBLEM STATEMENT

Critical Context

South Africa's Transit & GBV Safety Crisis

Vulnerability after dark, delayed emergency response, and evidence destruction.

1 in 3 Women

South Africa suffers from pandemic-level rates of Gender-Based Violence (GBV). Victims frequently travel alone and lack safe, discreet intervention channels.

Night Transit Risk

Millions of commuters rely on e-hailing (Uber/Bolt) and minibus taxis. Unauthorized route deviations, unregistered drivers, and dark transit corridors pose severe risks.

45+ Min 10111 Delay

Calling 10111 or 112 during an active attack or robbery is dangerous, slow, and fails to transmit real-time satellite GPS coordinates to first responders.

Evidence Destruction

Perpetrators routinely seize, search, or destroy victim devices. Traditional photos and text messages are lost, undermining legal prosecution in courts.

SOLUTION ARCHITECTURE

✨ Core Solution

Introducing Phola: An Autonomous Digital Bodyguard

A multi-layered safety shield designed specifically for South African commuters.



SafeRide Watchdog

Monitors trips autonomously. If your driver deviates from the expected route or stops in an unlit corridor, Phola automatically checks in and pings your emergency circle.



Native Police Navigation

1-tap routing directly inside Phola to the nearest SAPS police station or trauma clinic. Continuous turn-by-turn guidance keeps the user protected inside the app.



Silent 1-Tap SOS

Instant multi-channel panic alerting SAPS (10111), Emergency Medical (112), and the National GBV Command Centre (0800 428 428) with live satellite coordinates.



Zero-Knowledge Vault

Audio notes, photo logs, and incident diaries are encrypted on-device via Web Crypto SHA-256 and AES-GCM before syncing to the cloud. Tamper-evident and secure.



Stealth Decoy Exit

Under physical threat or phone inspection, 1 tap on "Quick Exit" instantly wipes DOM trace states and presents an innocent local weather forecast within 50ms.

ZA

11 Official Languages

Full native accessibility in isiZulu, isiXhosa, Afrikaans, Sepedi, Setswana, Sesotho, Xitsonga, siSwati, Tshivenda, isiNdebele, and English.

USER STORY

📖 Persona Journey

Thandi's Midnight Ride: Real-World Experience

How Phola protects a student returning home late from campus.

11:15 PM — **Booking Ride & Activating Watchdog**

Thandi leaves the university library and requests an e-hailing cab. She taps **"Start SafeRide"** in Phola. The app locks her vehicle notes and initiates background GPS tracking.

11:28 PM — **Route Deviation Anomaly Detected**

The vehicle abruptly takes an unlit access bypass away from her residential address. Phola detects the corridor divergence, alerts Thandi, and silently transmits her live GPS pin to her sister.

11:30 PM — **1-Tap Safe Haven: Go to Nearest SAPS Station**

Feeling threatened, Thandi taps **"Go to Nearest Police Station"**. Phola automatically plots an in-app high-contrast route to Brixton SAPS (1.8 km away) with live turn-by-turn instruction banners.

11:36 PM — **Safe Arrival & Cryptographic Evidence Preserved**

Driver pulls into the illuminated police station gate. Thandi exits safely. The entire telemetry log and route record remain encrypted in her Zero-Knowledge incident vault.

PRODUCT WALKTHROUGH

Feature Suite

Interactive Core Features & Interface

Engineered for high stress situations: high contrast, zero clutter, instant response.

🗺️ In-App SOS Map

- Continuous high-accuracy GPS tracking with user heading arrow.
- 500m pulse emergency radius circle.
- Multi-tier resilient routing: Mapbox API ➡️ OSRM ➡️ Straight-line vector.
- 1-tap route directly to nearest SAPS police station.

🔐 Biometric & PIN Security

- Native Apple Face ID (iOS) & Android Fingerprint sensor support.
- Auto-prompts on app resume or next sign-in.
- Salted SHA-256 PIN fallback.
- Smooth unblocked sign-out for all users.

🗄️ Zero-Knowledge Vault

- Client-side AES-GCM encrypted notes, photos, and voice notes.
- Encrypted before cloud sync—even database admins cannot read files.
- 1-tap data deletion complying with POPIA legislation.

🌦️ Stealth Weather Decoy

- Fixed top-corner Quick Exit accessible from any screen.
- Instantly unmounts emergency telemetry and renders realistic weather.
- Protects victims facing physical coercion or phone inspection.

ZA Multilingual Localization

- Minimal header pill for seamless language toggling.
- Full translation coverage across all 11 official South African languages.
- Eliminates linguistic barriers during high-stress emergencies.

⚡ Enterprise Sign-In / Sign-Up

- Unified segmented pill control.
- Google 1-Tap OAuth & live password strength meter.
- Pre-registered emergency contact setup during onboarding.
- POPIA consent agreement.

TECHNICAL ARCHITECTURE

System Design

Production Technology Stack & Resilience

Modern, hybrid, and fault-tolerant architecture built for offline durability.

Frontend & Hybrid

Next.js 16 + Capacitor 5

• React 19 with Turbopack for sub-second build and execution.
• Capacitor 5 Android & iOS bridges compile native native packages.
• Tailwind CSS + Radix UI with South African design palette (`#0D6E6E`).
• Progressive Web App (PWA) with offline caching.

Backend & Storage

Supabase PostgreSQL

• Row-Level Security (RLS) enforcing strict user-isolated boundaries.
• Supabase Auth JWT with automatic stale token invalidation.
• Cloudflare Workers edge proxying for low-latency dispatch.
• Realtime subscriptions for trip tracking.

GIS & Routing

Resilient Geolocation Engine

• Continuous GPS heading & speed telemetry via Geolocation API.
• Mapbox Directions API with automatic fallback to OSRM nodes.
• Offline straight-line compass fallback if network drops completely.
• Turf.js spatial calculation for nearest police station discovery.

Zero External Map Dependency
Unlike competing panic buttons that divert to Google Maps, Phola navigates entirely within the secure app container.

Containerized Safety

SECURITY & COMPLIANCE

SSDLC Certified

Secure SDLC & POPIA Compliance

Adherence to South African Privacy Law and Defense-in-Depth Cryptography.

1. Requirements

• Data Minimization: Zero passive background tracking without active trip.
• Explicit POPIA consent gathered on sign-up.
• Right to be Forgotten: 1-tap complete data purge in Settings.

2. Architecture

• STRIDE Threat Modeling applied across all data flows.
• Zero-Knowledge cryptography: Web Crypto AES-GCM client-side encryption.
• Row-Level Security (RLS) on all database tables.

3. Implementation

• Salted SHA-256 PIN hashing (`${userId}:phola-pin-salt:${pin}`).
• WebAuthn Passkeys & Biometrics (`PublicKeyCredential`).
• Strict TypeScript with zero any-casts in crypto routines.

4. Verification

• Automated SAST & type-checking (`tsc --noEmit`).
• Next.js static page bundle auditing (28 routes verified).
• Native Android APK compiled with Gradle 8 and Java 17.

BUSINESS MODEL

💼 Strategy

Lean Business Canvas Overview

Viable, scalable, and socially impactful business strategy.

Problem & Segments	Value Proposition & Solution	Channels & Advantage	Revenue Streams	Cost Structure
Problem: • 1 in 3 women face GBV • Night e-hailing risks • 45+ min 10111 response delay • Evidence destruction **Customers:** • Commuters & students • Night healthcare staff • Campuses & CPFs	**Unique Value Prop:** "Your autonomous travel watchdog and GBV companion with in-app police navigation and zero-knowledge vaults." **Solution:** • SafeRide trip watchdog • In-app turn-by-turn routing • 1–Tap SOS to 10111/112 • Stealth weather exit • 11 official SA languages	**Channels:** • PWA & Play Store • University campus drives • GBV shelter networks (POWA) • Community policing (CPF) **Unfair Advantage:** • Native in-app navigation • Zero-Knowledge crypto • 11 Official SA languages	• **Free Tier:** Public safety & SOS (always free) • **Phola Plus:** R39/mo armed response integration • **B2B Campus:** R15/student/mo campus safety portal • **CSI & Grants:** GBV justice sponsorships	• Next.js hosting & Supabase DB • Mapbox & OSRM routing APIs • SMS/WhatsApp emergency gateways • POPIA audits & maintenance

MARKET OPPORTUNITY

📈 Growth Strategy

South African Market Scale & Growth Path

Targeting 15 million daily commuters and 2.5 million higher-education students.

15M+

DAILY COMMUTERS

South Africans traveling via minibus taxis, e-hailing, and metro trains daily, with over 40% traveling during dark early-morning or late-night hours.

2.5M

TERTIARY STUDENTS

Enrolled across 26 public universities and 50 TVET colleges. Students represent our primary launchpad through institutional campus partnerships.

< 3s

EMERGENCY LATENCY

Benchmark latency from user 1–tap SOS activation to emergency SMS transmission with live GPS coordinates to personal contacts.

🤝 Strategic Partnership Strategy

1. University Campuses: Integration into university security operations (Wits, UJ, UCT, UP) for perimeter and transit monitoring.

2. Armed Response Networks: API sync with private security providers (Fidelity, ADT) for premium rapid dispatch.

3. GBV Support Shelters: Partnerships with organizations like POWA and TEARS Foundation for verified trauma referrals.

JUDGE VERIFICATION

✅ Verified Links

Official Submission Deliverables & Live Access

All materials required for judging are live, verified, and ready to inspect.

🌐 Live Web Application (PWA)

Hosted and deployed live on Vercel edge infrastructure:

```
https://phola-beta.vercel.app
```

Compatible with modern mobile and desktop browsers. Supports progressive installation and WebAuthn biometrics.

📦 Android APK Package

Compiled native debug APK ready for Android testing:

```
phola-debug.apk (3.64 MB)
```

Built with Gradle 8 & Java 17. Stored directly in the project root directory and available for immediate installation.

💻 Open Source Git Repository

Complete clean commit history on branch `main`:

```
https://github.com/jackyphuti/Phola
```

Includes full source code, Next.js routes, Capacitor mobile bridges, and comprehensive documentation in `PHOLA_SUBMISSION_PACK.md`.

🔑 Judge Demo Credentials

Live account created and verified for the evaluation panel:

```
Email: jackympoka@gmail.com
Password: Mpokabetty221@
```

Video demo will be uploaded to Google Drive following physical recording.

EVALUATION GUIDE

🧪 Test Protocol

Judge 5-Minute Evaluation Walkthrough

Step-by-step instructions to test Phola's breakthrough features.

Step 1

1. Sign In & Languages

• Open `https://phola-beta.vercel.app` or launch the APK.
• Tap the top language pill to switch to **isiZulu**, **isiXhosa**, or **Afrikaans**.
• Log in with **jackympoka@gmail.com** / **Mpokabetty221@**.

Step 2 (Key Test)

2. In-App Navigation

• Tap **Safe Ride** on the bottom navigation.
• Tap **"Go to Nearest Police Station (SAPS)"**.
• Watch the high-contrast emerald route render with live turn-by-turn guidance banners without opening external maps.

Step 3

3. 1-Tap SOS Panic

• Tap the central red **SOS / Panic** button.
• Verify instant 1-tap links for SAPS (10111), Emergency Medical (112), and the National GBV Command Centre.

Step 4

4. Biometrics / PIN

• Navigate to **Settings** (`/settings`).
• Enable **Face ID** (iOS) or **Fingerprint** (Android) or set a 4-digit PIN.
• Sign out and sign back in to observe automatic biometric authentication.

Step 5

5. Stealth Quick Exit

• From any screen, tap **Quick Exit** in the upper right corner.
• Observe the immediate <50ms teardown into an innocent, fully realistic weather forecast screen.

ZA

Phola Safety Network

"Phola" is a South African Nguni word meaning *"relax, be at ease, be safe"*.

Built by @Jacky Mpoka for the safety of all South Africans.